8*10/100/1000Base-T + 4*1G/2.5G/10G SFP+

Layer 2+ Managed Ethernet Switch

LT-S8312XG-4XS8GT

Overview

SmartByte LT-S8312XG-4XS8GT is a Layer 2+ full gigabit managed switch, which features 8-port 10/100/1000Base-T RJ45 and 4*1G/2.5G/10G SFP+ ports. It supports Layer 3 static routing with a high-performance switching architecture and offers wire-speed transportation capability for delivering a cost-effective 10G Ethernet solution to meet the converged applications for enterprise customer's networks. More and more corporations are adapting new IT technologies over the network such as voice over IP, video conference to improve productivity and save operation expenditure. The LT-S8312XG-4XS8GT series provide comprehensive end to end QoS as well as flexible and rich management, security settings, which can meet the high speed, secure and smart requirement for small and medium-size enterprise networks based on cost effective price.

Picture



Features

- 4*1G/2.5G/10G SFP+ slots uplink, 8 *10/100/1000BASE-T ports downlink, 1*RJ45 console interface for switch basic management and setup.
- **Hardware:** Marvell AC5 series 1.2GHz CPU, Marvell AC5 series switch chipsets, 512MB RAM and 512MB flash
- **Layer 3 features:** IPv4/IPv6 static routing, DHCPv4/ DHCPv6 client, DHCPv4 relay, DHCP relay.
- **Port features:** Supports port aggregation, IEEE802.3x flow control, port isolation, port mirroring, loop detection and broadcast storm suppression
- **VLAN:** Supports IEEE802.1Q tagged VLAN, mac based VLAN, IP based VLAN, protocol based VLAN, voice VLAN and private VLAN.
- **QinQ:** Supports port-based, VLAN-based and flow-based QinQ
- Supports **GVRP/LLDP/SPAN/RSPAN/DHCPv4 snooping**
- **Layer 2 ring network protocol:** Supports STP, RSTP, MSTP, G.8032 ERPS protocol, single ring, sub ring
- **Multicast:** IGMP snooping, IGMP v1, v2, v3, PIM-SM
- **QoS:** QoS class, remarking, Supports SP, WRR queue scheduling, Ingress/Egress port-based rate-limit and policy-based QoS
- **Security:** Supports ACL, 802.1x， port-security, IP source guard, ARP-check, DAI, anti-gateway ARP spoofing, Ddos attack protection, telnet/SSH/HTTP/HTTPS user access control and AAA.
- **Reliability:** Loop detect based on port or VLAN, the Ethernet in the first mile 802.3ah
- **Management:** Supports IPv4/IPv6 dual stack management, system log management, user management, login service management, firmware management, web management, CLI management, fault detection, DNS, NTP, sFlow, SNMP, MQTT, RMON and dual partition management.
- **Application protocol:** Supports telnet server, telnet client, SCP, SFTP, SSH server, TACACS and TFTP client

Technical specification

Model No.	LT-S8312XG-4XS8GT	
Interface	Fiber ports	Copper RJ45 ports
	4	8
Ethernet	8*10/100/1000Base-T RJ45 4*1G/2.5G/10GBase-X SFP+	
CPU	Marvell AC5 series 1.2GHz	
Size of RAM	512MB	
Size of Flash	512MB	
Switch chip model	Marvell AC5 series	
Management port	1*RJ45 Console port (115200,8,N,1)	
Standard	IEEE 802.3 10Base-T Ethernet IEEE 802.3u 100Base-TX Ethernet IEEE 802.3ab 1000Base-T Ethernet IEEE 802.3z 1000Base-X Ethernet IEEE 802.3ae 10Gigabit Ethernet IEEE802.3x Flow Control and Back Pressure IEEE 802.1D Spanning Tree Protocol IEEE 802.1w Rapid Spanning Tree Protocol IEEE 802.1Q VLAN ITU-T G.8032 ERPS IEEE 802.1X Port Authentication Network Control IEEE 802.1ab LLDP IEEE 802.3ad LACP	
LED Indicators		
P (Power indicator)	Off: the device is power off or failed	On: the device power on is normal
S (System status indicator)	Blinking: device initialization	On: device on normal operation
1-8 (Copper ports indicators) Green	Off: ports link down	
	On: ports link up	
	Blinking: data on TX/RX	
1-8 (Port speed indicator) Yellow	Off: Port speed on 10/100M	
	On: Port speed on 1000M	
9-12 (Fiber ports indicators)	Off: ports link down	

Green	On: ports link up
	Blinking: data on TX/RX
Power parameter	
Input voltage	AC 100 ~ 240V input
Total power consumption	Full loading ≤16W
Over-current protection	Support
Layer 2 switching	
Port aggregation	Support GE port aggregation
	Support 2.5GE aggregation
	Support 10GE aggregation
	Support static aggregation
	Support LACP dynamic aggregation
	Up to 64 aggregation groups and up to 8 ports per group
Port features	Support IEEE802.3x flow control
	Support interface counters
	Support port isolation
	Port mirroring (One-to-One, Many-to-One)
	Support loop detection (Port-based; VLAN-based)
	Support broadcast storm suppression (broadcast; unknown multicast; unknown unicast)
VLAN	IEEE802.1Q tagged VLAN
	Mac based VLAN
	IP based VLAN
	Protocol based VLAN
	Voice VLAN, specially divide VLAN for user voice data flow
	Private VLAN, divide a VLAN broadcast domain into multiple subdomains to form a double-layer VLAN structure
	VLAN range: 1-4094
	Support access mode/trunk mode/hybrid mode
Mac-address table management	Support static Mac-address management
	Support dynamic Mac-address Management

	Support filtering Mac-address
	Support MAC limit based on port and VLAN
	Support MAC flapping based on port and VLAN
	16K Mac-address table size
GVRP	Normal mode
	Fixed mode
	Forbidden mode
QinQ	Port-based QinQ
	VLAN-based QinQ
	VLAN mapping
	Flow-based QinQ
LLDP	Link Layer Discovery Protocol
	Med
SPAN	Number of Sessions: 7
	Many-to-one
RSPAN	Remote Switch Port Analyzer
MSTP	802.1D STP
	802.1W RSTP
	802.1S MSTP
	BPDU Guard
	Number of Single Ring Device Connections: 20
ERPS G.8032	Major ring
	Sub ring
	Recovery time ≤20ms
	Number of Single Ring Device Connections: No limit
QoS	QoS Class, Remarking
	Support SP, WRR Queue Scheduling
	Ingress Port-based Rate-limit
	Egress Port-based Rate-limit
	Egress port-queue shaper

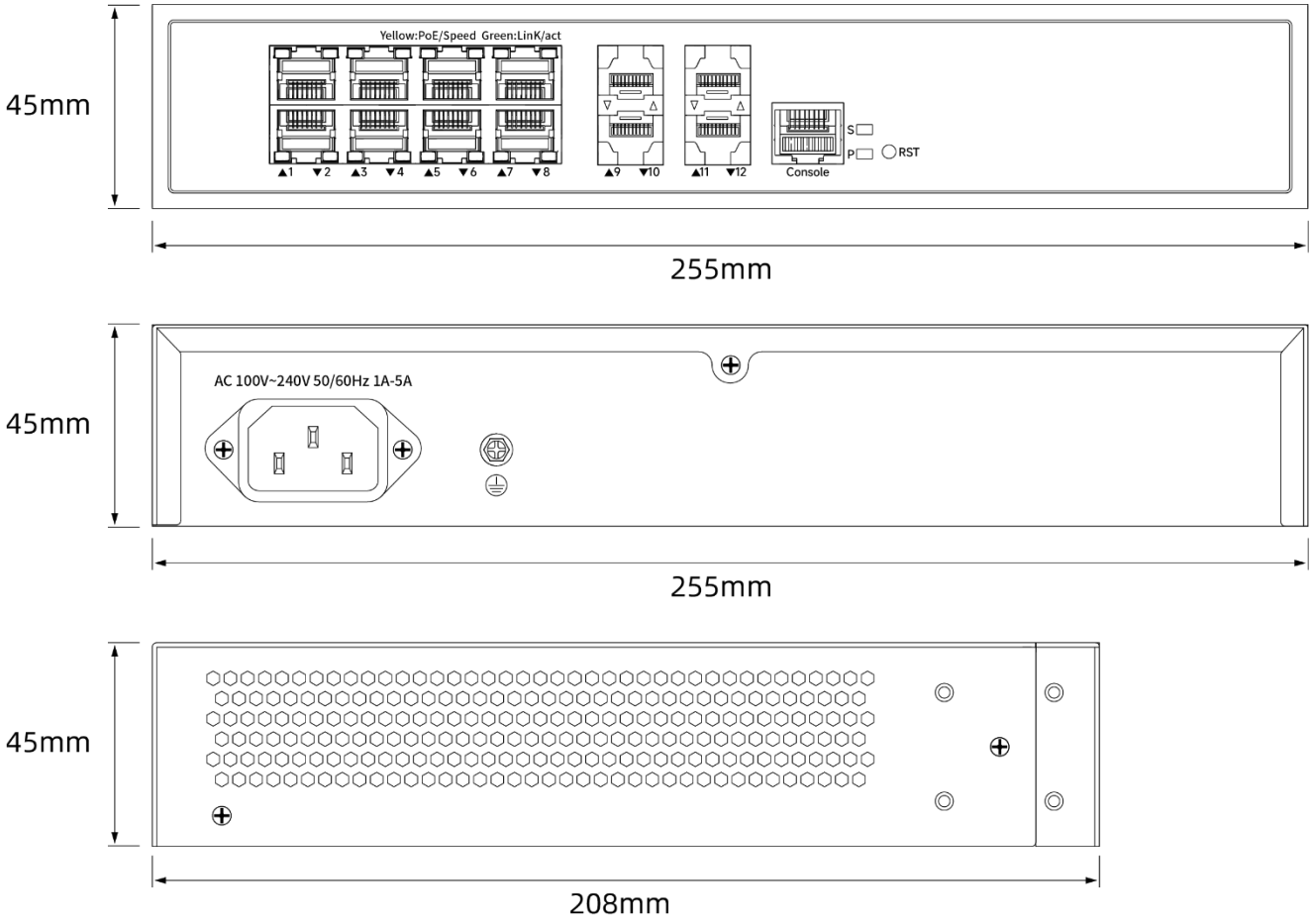
	Policy-based QoS
	Port Queue Numbers: 8
DHCPv4 Snooping	DHCP snooping trust port
	DHCP snooping option-82
	DHCP snooping circuit-id and remote-id customized
L2 Multicast	
IGMP	IGMP v1, v2, v3
IGMP Snooping	Internet Group Management Protocol Snooping
	Fast Leave
	Multicast Table Size: 1K
PIM-SM	Protocol Independent Multicast-Sparse Mode
Security features	
ACL	IP Standard ACL IP Extended ACL MAC extend ACL IPv6 ACL Time Range ACL apply on VLAN ACL entries, include input and output
802.1x	Port access control Mac-address access control RADIUS server
Port-security	Port-security
IP source guard	IP Source Guard, Port-based IP/Mac Binding Access
ARP-check	Support ARP-check and ARP packet filtering for illegal users
DAI	Dynamic ARP Inspection
Anti-gateway ARP Spoofing	Prevent Gateway ARP Spoofing
Ddos	Ddos attack protection
Access control	Support Telnet/SSH/HTTP/HTTPS user access control
AAA	RADIUS authentication access switch
	TACACS authentication access switch
	Authorization
Reliability	

Loop Detect	Loop detect, based on port or VLAN
EFM	The Ethernet in the first mile, 802.3ah
Management & Maintenance functions	
Dual Stack	IPv4 and IPv6 Dual Stack Management
System Log Management	Syslog Level 7 to 0 Corresponding Debug/Info/Notice/Warning/ Err/ Critical/ Alert
	Support connecting up to 3 Servers
	For IPv4 and IPv6
User Management	Support password protection, support MD5/SHA256/SHA512 algorithm
	Local Management of User Privilege
	Remote authorization by RADIUS or TARCAS
	Enable privilege protection
	User password complexity requirements
	User login fail protection
	User Privilege Level 0 to 15
Login Service Management	Up to 8 users online simultaneously
	Access control based on IPv4 or IPv6 ACL
Firmware Management	Firmware upgraded by WEB
	Firmware upgraded by CLI
	Dual partition
Web Management	Web Management
	Support HTTP V1.1
	Support HTTPS
	For IPv4 and IPv6
CLI management	Console/Telnet command line management
Fault Detection	Ping/Traceroute, For IPv4 and IPv6
	Optical transceiver DDM
	One click fault information collection
	Cable detection
DNS	Domain Name System

NTP	Network Time Protocol, For IPv4 and IPv6
	Support domain name configuration
Configuration Import /Export	FTP/TFTP based remote configuration import and export
sFlow	Sampled flow
SNMP	Version SNMPV2C/V3
MQTT	Message Queuing Telemetry Transport
RMON	Support RMON (Remote Monitoring) alarm
Dual partition management	Support dual partition switching
Application Protocol Functions	
Telnet server	Support telnet server for IPv4 and IPv6
Telnet client	Support telnet client for IPv4 and IPv6
SCP (Secure Copy Protocol)	SCP client/server for IPv4 and IPv6
SFTP	sftp client/server for IPv4 and IPv6
SSH server	Support login based on user password-SSH2, support IPv4, IPv6
	Support public key-based login
TACACS	Support TACACS (the terminal access controller accesses the control system)
TFTP	Support TFTP Client
Layer 3 features	
Static Route	Support VLAN interface and route port
	IPv4 static routing table entry configuration
	ARP dynamic learning and static configuration
	IPv6 static routing table entry configuration
	Neighbor discovery and static configuration
	Routing prefix table size 4k for IPv4, for IPv6 divide by 4
DHCPv4 Client	Dynamic host configuration protocol for IPv4
DHCPv6 Client	Dynamic host configuration protocol for IPv6
DHCPv4 Relay	DHCP Relay for IPv4
DHCPv4 Server	DHCP Server for IPv4
Switching features	
Switching capacity	96Gbps

Packet forwarding rate	142.8Mpps
MAC address table	16K
VLAN	4K
Buffer	12Mbit
Forwarding delay	<5us
Jumbo Frame	Support 10Kbytes
MDX/MIDX	Support
Watchdog	Support
Mechanical structure	
Installation method	Desktop; Rack-mount
Dimension(W*D*H) mm	255*208*45mm
Weight	0.66 kg
Operating environment	
Operating temperature	0℃~ +50℃
Storage/transportation temperature	-10℃~ +70℃
Relative humidity	Operation humidity: 10%-90%RH Storage humidity: 5%-95%RH
Protection	Surge protection of power: 4KV
	Surge protection of Ethernet ports: 4KV
	ESD: Contact 6kV, Air 8kV
Certification	CCC/CE/UKCA/RoHS/REACH/FCC
MTBF	> 100,000 hours
Warranty	3 years

Structure diagram



Order information

Model	Description
LT-S8312XG-4XS8GT	Layer 2+ managed Ethernet switch with 4*1G/2.5G/10G SFP+ slots and 8*10/100/1000M RJ45 ports, internal power supply, AC 100 ~ 240V input